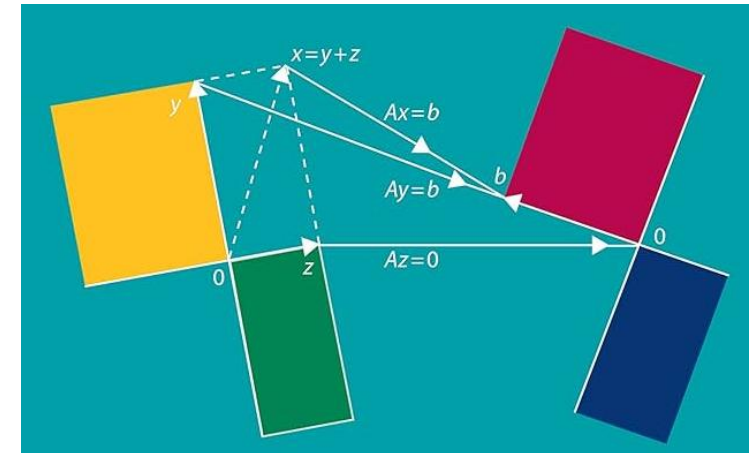


Transpose and Determinants of a Matrix

(Linear Algebra)

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Transpose of a Matrix

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \text{ then the transpose of this matrix is } \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}$$

A column vector such as $\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ can be written as the transpose of a row vector, that

is $\mathbf{x} = (x_1 \ \cdots \ x_n)^T$. This approach of writing a column vector $\mathbf{x}$ as the transpose of a row vector saves space. In this case, column vector $\mathbf{x}$ takes up n lines while $\mathbf{x} = (x_1 \ \cdots \ x_n)^T$ can be written on a single line.

The Transpose of A

We need one more matrix, and fortunately it is much simpler than the inverse. It is the “*transpose*” of A , which is denoted by A^T . The columns of A^T are the rows of A .

When A is an m by n matrix, the transpose is n by m . Then 2 by 3 becomes 3 by 2.

Transpose If $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 4 \end{bmatrix}$ then $A^T = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 4 \end{bmatrix}$.

You can write the rows of A into the columns of A^T . Or you can write the columns of A into the rows of A^T . The matrix “flips over” its main diagonal. The entry in row i , column j of A^T comes from row j , column i of the original A :

Rows of A become columns of A^T

$$(A^T)_{ij} = A_{ji}$$

The transpose of a lower triangular matrix is upper triangular. The transpose of A^T is A .

Properties of Transpose Matrix

Let A and B be matrices of appropriate size so that the operations below can be carried out. We have the following properties (k is a scalar):

$$(a) (A^T)^T = A$$

$$(b) (kA)^T = kA^T$$

$$(c) (A + B)^T = A^T + B^T$$

$$(d) (AB)^T = B^T A^T$$

The rules for transposes are very direct. We can transpose $A + B$ to get $(A + B)^T$. Or we can transpose A and B separately, and then add $A^T + B^T$ —with the same result.

The serious questions are about the transposes of a product AB and an inverse A^{-1} :

Sum	The transpose of $A + B$ is $A^T + B^T$.	(1)
Product	The transpose of AB is $(AB)^T = B^T A^T$.	(2)
Inverse	The transpose of A^{-1} is $(A^{-1})^T = (A^T)^{-1}$.	(3)

Proof

$$\begin{aligned}(\mathbf{A} + \mathbf{B})^T &= \left[\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{pmatrix} \right]^T \\&= \begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \cdots & a_{2n} + b_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \cdots & a_{mn} + b_{mn} \end{pmatrix}^T \quad \left[\begin{array}{l} \text{adding the} \\ \text{corresponding} \\ \text{entries} \end{array} \right] \\&= \begin{pmatrix} a_{11} + b_{11} & a_{21} + b_{21} & \cdots & a_{m1} + b_{m1} \\ a_{12} + b_{12} & a_{22} + b_{22} & \cdots & a_{m2} + b_{m2} \\ \vdots & \vdots & \vdots & \vdots \\ a_{1n} + b_{1n} & a_{2n} + b_{2n} & \cdots & a_{mn} + b_{mn} \end{pmatrix} \quad \left[\begin{array}{l} \text{taking the transpose,} \\ \text{that is interchanging} \\ \text{rows and columns} \end{array} \right] \\&= \underbrace{\begin{pmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \vdots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{pmatrix}}_{=\mathbf{A}^T} + \underbrace{\begin{pmatrix} b_{11} & b_{21} & \cdots & b_{m1} \\ b_{12} & b_{22} & \cdots & b_{m2} \\ \vdots & \vdots & \vdots & \vdots \\ b_{1n} & b_{2n} & \cdots & b_{mn} \end{pmatrix}}_{=\mathbf{B}^T} = \mathbf{A}^T + \mathbf{B}^T\end{aligned}$$

Let $\mathbf{A}$ be an invertible (non-singular) matrix then the transpose of the matrix, $\mathbf{A}^T$, is also invertible and $(\mathbf{A}^T)^{-1} = (\mathbf{A}^{-1})^T$.

- For an invertible matrix, you can change the order of the inverse and transpose.

$$(\mathbf{A}^{-1})^T \mathbf{A}^T = \mathbf{I} \text{ and } \mathbf{A}^T (\mathbf{A}^{-1})^T = \mathbf{I}$$

Examining the first of these $(\mathbf{A}^{-1})^T \mathbf{A}^T$ we have

$$\begin{aligned} (\mathbf{A}^{-1})^T \mathbf{A}^T &= (\mathbf{A}\mathbf{A}^{-1})^T && \text{[using Theorem } \mathbf{Y}^T \mathbf{X}^T = (\mathbf{X}\mathbf{Y})^T \\ &= \mathbf{I}^T && \text{[remember } \mathbf{A}\mathbf{A}^{-1} = \mathbf{I}] \\ &= \mathbf{I} && \text{[remember } \mathbf{I}^T = \mathbf{I} \text{ because } \mathbf{I} \text{ is the identity}] \end{aligned}$$

Transpose of inverse $A^{-1}A = I$ is transposed to $A^T(A^{-1})^T = I$. (7)

Similarly $AA^{-1} = I$ leads to $(A^{-1})^TA^T = I$. We can invert the transpose or we can transpose the inverse. Notice especially: A^T *is invertible exactly when A is invertible*.

The inverse of $A = \begin{bmatrix} 1 & 0 \\ 6 & 1 \end{bmatrix}$ is $A^{-1} = \begin{bmatrix} 1 & 0 \\ -6 & 1 \end{bmatrix}$. The transpose is $A^T = \begin{bmatrix} 1 & 6 \\ 0 & 1 \end{bmatrix}$.

$(A^{-1})^T$ and $(A^T)^{-1}$ are both equal to $\begin{bmatrix} 1 & -6 \\ 0 & 1 \end{bmatrix}$.

The Meaning of Inner Products

The dot product (inner product) of $\mathbf{x}$ and $\mathbf{y}$ is the sum of numbers $x_i y_i$. Now we have a better way to write $\mathbf{x} \cdot \mathbf{y}$, without using that unprofessional dot. Use matrix notation $\mathbf{x}^T \mathbf{y}$:

$\mathbf{T}$ is inside *The dot product or inner product is $\mathbf{x}^T \mathbf{y}$* $(1 \times n)(n \times 1) = 1 \times 1$

$\mathbf{T}$ is outside *The rank one product or outer product is $\mathbf{x} \mathbf{y}^T$* $(n \times 1)(1 \times n) = n \times n$

Symmetric Matrices

For a *symmetric matrix*, transposing A to A^T produces no change. In this case A^T equals A . Its (j, i) entry across the main diagonal equals its (i, j) entry. In my opinion, these are the most important matrices of all. We give symmetric matrices the special letter S .

A symmetric matrix has $S^T = S$. This means that every $s_{ji} = s_{ij}$.

Symmetric matrices $S = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix} = S^T$ and $D = \begin{bmatrix} 1 & 0 \\ 0 & 10 \end{bmatrix} = D^T$.

The inverse of a symmetric matrix is a symmetric matrix. The transpose of S^{-1} is $(S^{-1})^T = (S^T)^{-1} = S^{-1}$. When S is invertible, this says that S^{-1} is symmetric.

Choose any matrix A , probably rectangular. Multiply A^T times A . Then the product $S = A^T A$ is automatically a square symmetric matrix :

The transpose of $A^T A$ is $A^T(A^T)^T$ which is $A^T A$ again. (10)

The matrix AA^T is also symmetric. (The shapes of A and A^T allow multiplication.) But AA^T is a different matrix from $A^T A$. In our experience, most scientific problems that start with a rectangular matrix A end up with $A^T A$ or AA^T or both. As in least squares.

Example 3 Multiply $A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$ times $A^T = \begin{bmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \end{bmatrix}$ in both orders.

$AA^T = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ and $A^T A = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$ are both symmetric matrices.

The product AA^T is m by m . In the opposite order, $A^T A$ is n by n . Both are symmetric, with positive diagonal (why?). But even if $m = n$, it is very likely that $A^T A \neq AA^T$.

Questions

- 1 Find A^T and A^{-1} and $(A^{-1})^T$ and $(A^T)^{-1}$ for $A = \begin{bmatrix} 1 & 0 \\ 9 & 3 \end{bmatrix}$ and $\begin{bmatrix} 1 & c \\ c & 0 \end{bmatrix}$.
- 2 Verify that $(AB)^T$ equals $B^T A^T$ but those are different from $A^T B^T$:

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \quad AB = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}.$$

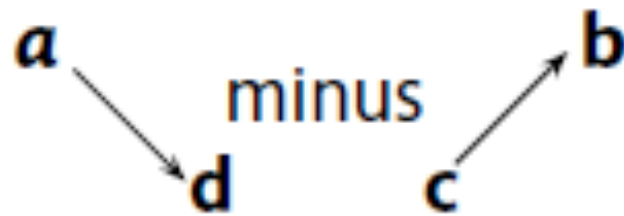
Show also that AA^T is different from $A^T A$. But both of those matrices are _____.

Determinant

- The *determinant* is a number associated with a square matrix
- Determinant of a square matrix is a number.
- Can use this value to establish whether the *matrix has an inverse or not*, as well as *finding whether the linear system has a unique solution*
- The determinant is notated as $\det(A)$ or $|A|$

Determinant of a 2 by 2 Matrix

$$\det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$



Determinant of a 2 by 2 Matrix

$$\det \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{1} \quad \det \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \mathbf{-1}$$

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \mathbf{ad - bc} \quad \det \begin{bmatrix} c & d \\ a & b \end{bmatrix} = \mathbf{bc - ad}$$

- Determinants reverse sign when two rows are exchanged (or two columns are exchanged).
- The matrix has no inverse when its determinant is zero.
- $\det A = 0$ means that the columns of A are not independent.

Determinant of a 3 by 3 Matrix

Start with the identity matrix ($\det I = 1$) and exchange two rows (then $\det = -1$). Exchanging again brings back $\det = +1$. You quickly have all six permutation matrices. Each row exchange will reverse the sign of the determinant (+1 to -1 or else -1 to +1):

$$\begin{array}{cccccc} \begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix} & \begin{bmatrix} & 1 & \\ 1 & & \\ & & 1 \end{bmatrix} & \begin{bmatrix} & & 1 \\ & 1 & \\ 1 & & \end{bmatrix} & \begin{bmatrix} & 1 & \\ & & 1 \\ 1 & & \end{bmatrix} & \begin{bmatrix} & & 1 \\ 1 & & \\ & 1 & \end{bmatrix} & \begin{bmatrix} 1 & & \\ & & 1 \\ & 1 & \end{bmatrix} \\ \det = +1 & -1 & +1 & -1 & +1 & -1 \end{array}$$

If I exchange two rows of A —say rows 1 and 2—then its determinant changes sign. This will carry over to all determinants: **Row exchange multiplies $\det A$ by -1 .**

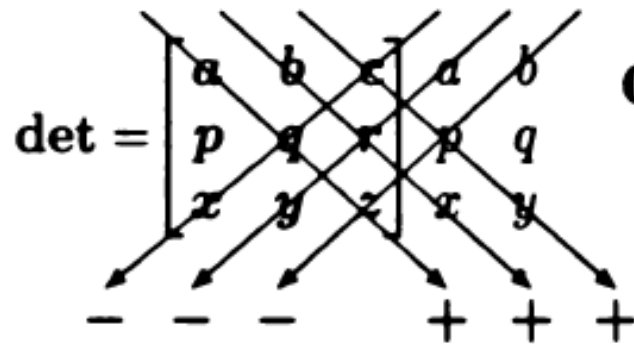
Determinant of a 3 by 3 Matrix

When you multiply a row by a number, this multiplies the determinant by that number. Suppose the three rows are $a\ b\ c$ and $p\ q\ r$ and $x\ y\ z$. Those nine numbers multiply ± 1 .

$$\begin{bmatrix} a & & \\ & q & \\ & & z \end{bmatrix} \begin{bmatrix} & b & \\ p & & \\ & & z \end{bmatrix} \begin{bmatrix} & & b \\ & r & \\ x & & \end{bmatrix} \begin{bmatrix} & & c \\ & q & \\ x & & \end{bmatrix} \begin{bmatrix} & & c \\ p & & \\ & y & \end{bmatrix} \begin{bmatrix} a & & \\ & & r \\ & y & \end{bmatrix}$$

$$\det = +aqz \quad -bpz \quad +brx \quad -cqx \quad +cpy \quad -ary$$

Finally we use the most powerful property we have. **The determinant of A is linear in each row separately.** As equation (4) will show, we can add those six determinants. To remember the plus and minus signs, I follow the arrows in this picture of the matrix.



Combine those 6 simple determinants into $\det A =$

$$\underline{+ aqz + brx + cpy - ary - bpz - cqx} \quad (1)$$

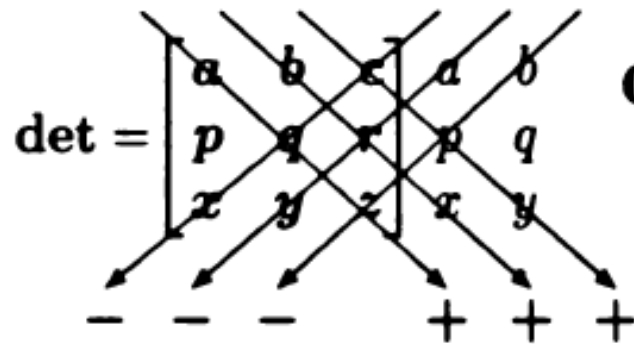
Determinant of a 3 by 3 Matrix

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$$\begin{bmatrix} a & & \\ & q & \\ & & z \end{bmatrix} \begin{bmatrix} & b & \\ p & & \\ & & z \end{bmatrix} \begin{bmatrix} & & b \\ & r & \\ x & & \end{bmatrix} \begin{bmatrix} & & c \\ & q & \\ x & & \end{bmatrix} \begin{bmatrix} & & c \\ p & & \\ & y & \end{bmatrix} \begin{bmatrix} a & & \\ & & r \\ & y & \end{bmatrix}$$

$$\det = +aqz \quad -bpz \quad +brx \quad -cqx \quad +cpy \quad -ary$$

Finally we use the most powerful property we have. **The determinant of A is linear in each row separately.** As equation (4) will show, we can add those six determinants. To remember the plus and minus signs, I follow the arrows in this picture of the matrix.



Combine those 6 simple determinants into $\det A =$

$$\underline{+ aqz + brx + cpy - ary - bpz - cqx} \quad (1)$$

Determinant of a 3 by 3 Matrix

$$\mathbf{A}_1 = [\mathbf{2}]$$

$$\det \mathbf{A}_1 = \mathbf{2}$$

$$\mathbf{A}_2 = \begin{bmatrix} \mathbf{2} & \mathbf{-1} \\ \mathbf{-1} & \mathbf{2} \end{bmatrix}$$

$$\det \mathbf{A}_2 = \mathbf{4} - \mathbf{1} = \mathbf{3}$$

$$\mathbf{A}_3 = \begin{bmatrix} \mathbf{2} & \mathbf{-1} & \mathbf{0} \\ \mathbf{-1} & \mathbf{2} & \mathbf{-1} \\ \mathbf{0} & \mathbf{-1} & \mathbf{2} \end{bmatrix}$$

$$\det \mathbf{A}_3 = \mathbf{8} - \mathbf{2} - \mathbf{2} = \mathbf{4}$$

Determinant of a 3 by 3 Matrix

Minors

$$\mathbf{A} = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$$

The *determinant* of the *remaining* matrix after *deleting* the *row* and *column* of an entry is called the **minor** of that entry.

Determinant of a 3 by 3 Matrix

Consider a square matrix $\mathbf{A}$. Let a_{ij} be the entry in the i th row and j th column of matrix $\mathbf{A}$. The **minor** M_{ij} of entry a_{ij} is the *determinant* of the remaining matrix after *deleting* the entries in the i th row and j th column.

$$M_{ij} = \det \left(\begin{array}{ccc|ccc} a_{11} & \cdots & \cdots & a_{1n} & & \\ \vdots & & & \vdots & & \\ \hline & & a_{ij} & & & \\ \hline a_{n1} & \cdots & \cdots & a_{nn} & & \end{array} \right)$$

j th column $\xrightarrow{\hspace{1.5cm}}$ i th row

Determinant of a 3 by 3 Matrix

Minor of entry a

$$\det \begin{pmatrix} e & f \\ h & i \end{pmatrix}$$

Minor of entry b

$$\det \begin{pmatrix} d & f \\ g & i \end{pmatrix}$$

Determinant of a 3 by 3 Matrix

Cofactor: Consider a square matrix A . Let a_{ij} be the entry in the i^{th} row and j^{th} column of matrix A . The cofactor c_{ij} of the entry a_{ij} is defined as

$$C_{ij} = (-1)^{i+j} M_{ij}$$

where M_{ij} is the minor of entry a_{ij} .

$$\begin{pmatrix} + & - & + & \dots \\ - & + & - & \dots \\ + & - & + & \dots \\ \vdots & \vdots & \vdots & \vdots \end{pmatrix}$$

Question

Determine the cofactor of 5 in $\begin{pmatrix} 3 & 5 & 7 \\ -1 & 2 & 3 \\ -4 & 4 & -9 \end{pmatrix}$

$$\mathbf{A} = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \quad \left[\text{remember that place signs are } \begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix} \right]$$

$$\det(\mathbf{A}) = a (\text{cofactor of } a) + b (\text{cofactor of } b) + c (\text{cofactor of } c)$$

$$\det(\mathbf{A}) = a \left[\det \begin{pmatrix} e & f \\ h & i \end{pmatrix} \right] - b \left[\det \begin{pmatrix} d & f \\ g & i \end{pmatrix} \right] + c \left[\det \begin{pmatrix} d & e \\ g & h \end{pmatrix} \right]$$

$$\mathbf{A} = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \quad \left[\text{remember that place signs are } \begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix} \right]$$

Middle Row

$$\det(\mathbf{A}) = d (\text{cofactor of } d) + e (\text{cofactor of } e) + f (\text{cofactor of } f)$$

Bottom Row

$$\det(\mathbf{A}) = g (\text{cofactor of } g) + h (\text{cofactor of } h) + i (\text{cofactor of } i)$$

$$\mathbf{A} = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \quad \left[\text{remember that place signs are } \begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix} \right]$$

We can also expand along any of the columns to find the determinant of $\mathbf{A}$.
The formula for expanding along the first column is

$$\det(\mathbf{A}) = a (\text{cofactor of } a) + d (\text{cofactor of } d) + g (\text{cofactor of } g)$$

If any of the rows or columns contain zeros then we choose to expand along that row or column because it simplifies the arithmetic.

Question

Compute the determinant of

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

Question

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by $T(\mathbf{x}) = \mathbf{A}\mathbf{x}$ where $\mathbf{A} = \begin{pmatrix} -1 & 5 & -2 \\ -6 & 6 & 0 \\ 3 & -7 & 1 \end{pmatrix}$. Find $\det(\mathbf{A})$.

Finding inverse of a Matrix

$$\det(\mathbf{A}) = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in} = \sum_{k=1}^n a_{ik}C_{ik}$$

Cofactor Matrix

Cofactor Matrix

Let $\mathbf{C}$ be the new matrix consisting of the cofactors of the general matrix $\mathbf{A}$. If

$$\mathbf{A} = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \quad \text{then} \quad \mathbf{C} = \begin{pmatrix} A & B & C \\ D & E & F \\ G & H & I \end{pmatrix}$$

where A is the cofactor of entry a , B is the cofactor of entry b , C is the cofactor of entry c ... The matrix $\mathbf{C}$ is called the **cofactor matrix** and it is used in finding the *inverse* of $\mathbf{A}$. Note that bold $\mathbf{C}$ represents the cofactor matrix and plain C is the cofactor of the entry c .

Question

Find the cofactor matrix **C** of

$$\mathbf{A} = \begin{pmatrix} 1 & -1 & 5 \\ 3 & 9 & 7 \\ -2 & 1 & 0 \end{pmatrix}$$

Adjoint

- We use the cofactor matrix to find the inverse of an invertible matrix. The final term we need to define is 'adjoint'.

- **Definition:**

Let $\mathbf{A}$ be a square matrix then the matrix consisting of the cofactors of each entry in $\mathbf{A}$ is called the cofactor matrix and is normally denoted by $\mathbf{C}$. The *transpose* of this cofactor matrix is called the **adjoint** of $\mathbf{A}$ and is denoted by $adj(\mathbf{A})$. That is

$$adj(\mathbf{A}) = \mathbf{C}^T$$

Inverse of a matrix

- Theorem:

If $\det(\mathbf{A}) \neq 0$ (not zero) then

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \text{adj}(\mathbf{A})$$

Inverse matrix formula

$$\mathbf{A}^{-1} = \mathbf{C}^T / \det \mathbf{A}$$

Question

Find the cofactor matrices $\mathbf{C}$ and $\mathbf{C}^T$ of $\mathbf{A} = \begin{pmatrix} 1 & 0 & 5 \\ -2 & 3 & 7 \\ 6 & -1 & 0 \end{pmatrix}$

Also determine $\mathbf{A}^{-1}$.

Properties of Determinants

1. The n by n identity matrix has $\det I = 1$.
2. Exchanging two rows of A reverses $\det A$ to $-\det A$.
3. If row 1 of A is a combination $cv + dw$, then add 2 determinants :

$$\det \begin{bmatrix} cv + dw \\ \text{row 2} \\ \vdots \\ \text{row } n \end{bmatrix} = c \det \begin{bmatrix} v \\ \text{row 2} \\ \vdots \\ \text{row } n \end{bmatrix} + d \det \begin{bmatrix} w \\ \text{row 2} \\ \vdots \\ \text{row } n \end{bmatrix}$$

Determinant of a Matrix is a linear function of each row separately

If a few elements of a row or column are expressed as a sum of terms, then the determinant can be expressed as a sum of two or more determinants.

$$\begin{vmatrix} a_1 + b_1 & a_2 + b_2 & a_3 + b_3 \\ c_1 & c_2 & c_3 \\ d_1 & d_2 & d_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ c_1 & c_2 & c_3 \\ d_1 & d_2 & d_3 \end{vmatrix} + \begin{vmatrix} b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \\ d_1 & d_2 & d_3 \end{vmatrix}$$

The elements of the first row represent the sum of terms, which can be split into two different determinants. Further, the new determinants also have the same second and third row, as the earlier determinant.

Left-Hand Side (Original Determinant):

$$\begin{vmatrix} 2+3 & 1+4 \\ 5 & 6 \end{vmatrix} = \begin{vmatrix} 5 & 5 \\ 5 & 6 \end{vmatrix}$$

- Calculate the value: $(5 \times 6) - (5 \times 5) = 30 - 25 = 5$

Right-Hand Side (Split Determinants):

Now, split Row 1 into two separate determinants while keeping Row 2 exactly the same: 

$$\begin{vmatrix} 2 & 1 \\ 5 & 6 \end{vmatrix} + \begin{vmatrix} 3 & 4 \\ 5 & 6 \end{vmatrix}$$

- Value of the first determinant: $(2 \times 6) - (1 \times 5) = 12 - 5 = 7$
- Value of the second determinant: $(3 \times 6) - (4 \times 5) = 18 - 20 = -2$
- Sum of both components: $7 + (-2) = 5$ 

Properties of Determinants

- Derived Properties

Property 4 If two rows of A are equal then $\det(A) = 0$.

Property 5 Subtracting a multiple of one row from another leaves $\det(A)$ unchanged.

We can see this is true in the 2×2 case by noting

$$\begin{aligned} \det \begin{pmatrix} a & b \\ c - ta & d - tb \end{pmatrix} &= \det \begin{pmatrix} a & b \\ c & d \end{pmatrix} - t \left(\det \begin{pmatrix} a & b \\ a & b \end{pmatrix} \right) \\ &= \det \begin{pmatrix} a & b \\ c & d \end{pmatrix}. \end{aligned}$$

Properties of Determinants

- Derived Properties

Property 6 A matrix with a row of zeros has $\det(A) = 0$.

If we multiply the row of zeros by t the matrix is unchanged, and so we must have $\det(A) = t \times \det(A)$ for all t , which is only true if $\det(A) = 0$.

Property 7 If A is triangular then $\det(A) = a_{11}a_{22} \cdots a_{nn}$, the product of the diagonal entries.

We can just use elimination to make all the non-diagonal entries 0, and then apply linearity n times.

Properties of Determinants

- Derived Properties

Property 8 If A is singular then $\det(A) = 0$. If A is invertible then $\det(A) \neq 0$.

We use elimination to go from A to U . Each step of elimination either leaves the determinant the same, or switches its sign. If A is singular then U has a zero on its diagonal, and so by property 7 the determinant is 0. On the other hand if A is invertible then U has pivots along its diagonal, and the product of non-zero terms is always non-zero.

Properties of Determinants

- Derived Properties

Property 8 If A is singular then $\det(A) = 0$. If A is invertible then $\det(A) \neq 0$.

The determinant of a square matrix tells us a lot. First of all, an invertible matrix has $\det A \neq 0$. A singular matrix has $\det A = 0$. When we come to eigenvalues λ and eigenvectors $\mathbf{x}$ with $A\mathbf{x} = \lambda\mathbf{x}$, we will write that eigenvalue equation as $(A - \lambda I)\mathbf{x} = \mathbf{0}$. This means that $A - \lambda I$ is singular and $\det(A - \lambda I) = 0$. We have an equation for λ .

Overall, the formulas are useful for small matrices and also for special matrices. And the properties of determinants can make those formulas simpler. If the matrix is triangular or diagonal, we just multiply the diagonal entries to find the determinant:

Triangular matrix
Diagonal matrix

$$\det \begin{bmatrix} a & b & c \\ & q & r \\ & & z \end{bmatrix} = \det \begin{bmatrix} a & & \\ & q & \\ & & z \end{bmatrix} = aqz \quad (1)$$

Properties of Determinants

- Derived Properties

Property 9 The determinant of AB is $\det(A)\det(B)$.

Property 10 The transpose A^T has the same determinant as A .

If we transpose A , the determinant formula gives the same result :

$$\text{Transpose the matrix} \qquad \det(A^T) = \det(A) \qquad (2)$$

If we multiply AB , we just multiply determinants (this is a wonderful fact):

$$\text{Multiply two matrices} \qquad \det(AB) = (\det A) (\det B) \qquad (3)$$

Cramer's Rule to solve $Ax = b$

- Cramer's rule gives us another way to solve $Ax = b$.
- Cramer's rule works better in theory than it works in practice.
- This is very useful to prove things about matrices and linear transformations.

$$\text{Key idea} \quad \mathbf{A} \mathbf{M}_1 = \mathbf{B}_1 \quad \begin{bmatrix} \mathbf{x}_1 & 0 & 0 \\ \mathbf{x}_2 & 1 & 0 \\ \mathbf{x}_3 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{b}_1 & a_{12} & a_{13} \\ \mathbf{b}_2 & a_{22} & a_{23} \\ \mathbf{b}_3 & a_{32} & a_{33} \end{bmatrix} = \mathbf{B}_1.$$

- Replace the first column of the identity matrix with the vector x

Cramer's Rule to solve $Ax = b$

This triangular M_1 has determinant x_1 .

Product rule $(\det A)(x_1) = \det B_1$ or $x_1 = \det B_1 / \det A.$

To find x_2 and B_2 , put the vectors x and b into the *second columns* of I and A :

$$\begin{array}{l} \text{Same idea} \\ AM_2 = B_2 \end{array} \quad \begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix} \begin{bmatrix} 1 & x_1 & 0 \\ 0 & x_2 & 0 \\ 0 & x_3 & 1 \end{bmatrix} = \begin{bmatrix} a_1 & b & a_3 \end{bmatrix} = B_2.$$

Cramer's Rule to solve $Ax = b$

CRAMER'S RULE If $\det A$ is not zero, $Ax = b$ is solved by determinants:

$$x_1 = \frac{\det B_1}{\det A} \quad x_2 = \frac{\det B_2}{\det A} \quad \dots \quad x_n = \frac{\det B_n}{\det A}$$

The matrix B_j has the j th column of A replaced by the vector b .

Question

Use Cramer's rule to solve for the vector $\mathbf{x}$:

$$\begin{pmatrix} -1 & 2 & -3 \\ 2 & 0 & 1 \\ 3 & -4 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$$

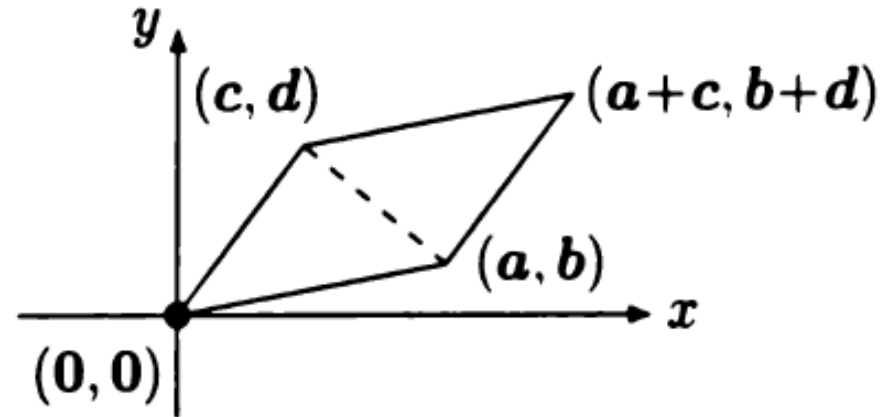
Areas and Volumes by Determinants

Determinants lead to good formulas for areas and volumes. The regions have straight sides but they are not square. A typical region is a parallelogram—or half a parallelogram which is a triangle. The problem is: **Find the area.** For a triangle, the area is $\frac{1}{2}bh$: half the base times the height. A parallelogram contains two triangles with equal area, so we omit the $\frac{1}{2}$. Then *parallelogram area = base times height*.

Those formulas are simple to remember. But they don't fit our problem, because we **are not given the base and height**. We only know **the positions of the corners**. For the triangle, suppose the corner points are $(0, 0)$ and (a, b) and (c, d) . For the parallelogram (twice as large) the fourth corner will be $(a+c, b+d)$. Knowing a, b, c, d , *what is the area?*

Areas and Volumes by Determinants

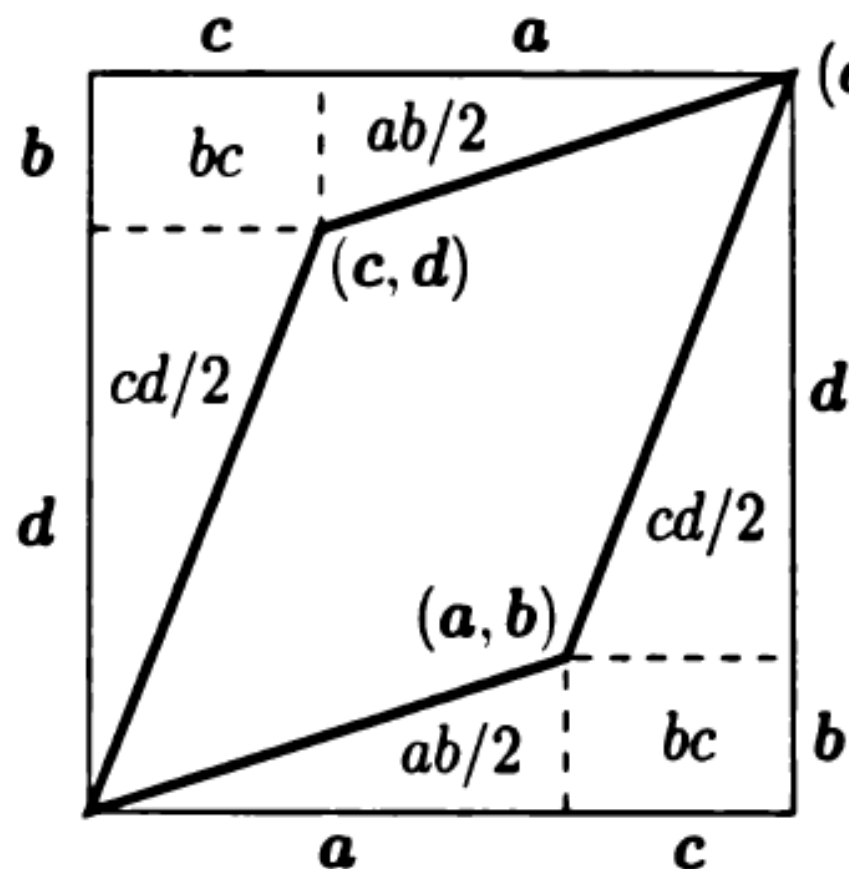
**Triangle and
parallelogram**



The base could be the lowest side and its length is $\sqrt{a^2 + b^2}$. To find the height, we could create a line down from (c, d) that is perpendicular to the baseline. The length h of that perpendicular line would involve more square roots. But the area itself does not involve square roots ! Its beautiful formula $ad - bc$ is simple for $a = d = 1, b = c = 0$ (a square).

$$\text{Area of parallelogram} = | \text{Determinant of matrix} | = \pm \begin{vmatrix} a & c \\ b & d \end{vmatrix} = |ad - bc|. \quad (1)$$

Our first effort stays in a plane. For this case we use geometry. Figure 5.1 shows how adding pieces to a parallelogram can produce a rectangle. When we subtract the areas of those six pieces, we arrive at the correct parallelogram area $ad - bc$ (no square roots). The picture is not very elegant, but in two dimensions it succeeds.



$(a + c, b + d) = (\text{base, height})$ of rectangle

Area of parallelogram

Rectangle minus 6 pieces

$$(a + c)(b + d) - 2bc - ab - cd \\ = ad - bc$$

Volume of a Box

$$\text{Volume of box} = | \text{Determinant of matrix} | = \pm \begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix}$$

Areas and Volumes by Determinants

- 1** A parallelogram in 2D starts from $(0, 0)$ with sides $\mathbf{e}_1 = (a, b)$ and $\mathbf{e}_2 = (c, d)$.
- 2** Area of the parallelogram = $\left| \text{Determinant of the matrix } E = \begin{bmatrix} \mathbf{e}_1 & \mathbf{e}_2 \end{bmatrix} \right| = |ad - bc|$.
- 3** A tilted box in 3D starts with three edges $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ out from $(0, 0, 0)$.
- 4** Volume of a tilted box = $|\text{Determinant of the } 3 \times 3 \text{ edge matrix } E|$.

Areas and Volumes by Determinants

A box in n dimensions has n edges $e_1, e_2, \dots, e_n$ going out from the origin. The parallelogram in two dimensions had two vectors $e_1 = (a, b)$ and $e_2 = (c, d)$. Those vectors e give two corners or n corners of the “box”. In the 2-dimensional picture, the fourth corner was $e_1 + e_2$. In the n -dimensional picture, the other corners of the box would be sums of the e ’s. *The box is totally determined by the n edges in the matrix E :*

$$\text{Edge matrix } E_2 = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \text{ and } E_n = \begin{bmatrix} | & & | \\ e_1 & \cdots & e_n \\ | & & | \end{bmatrix} \quad \begin{array}{l} \text{Columns =} \\ \text{Box edges} \end{array}$$

- If the rows of A are $a_1, a_2, \dots, a_n$, respectively, then $|\det(A)|$ is the n -dimensional volume (the generalization of area in R^2 and volume in R^3) of the parallelepiped having the vectors $a_1, a_2, \dots, a_n$ as adjacent sides.